

PAPER_236: UQFF Learning Assessment Evolution_B — Framework Meta-Assessment Advancement Score

Author: Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grok_share_8d951e12.txt second-pass — Doc 9) **Date:** March 2026 **Classification:** Novel — First Framework-Level Meta-Assessment Calculator in UQFF Pipeline **Status:** Proof-Quality Whitepaper **CP3 Class:** UQFFLearningAdvancementCalculator

Abstract

Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$; $[\text{SSq}] = 5.7\text{e-}1$; $H_{\text{SCm}} \approx 9.9\text{e-}1$; $U_{\text{UA}} \approx 1.0\text{e-}4$; $k_{\eta} = 1.0\text{e-}113$; $\beta_i \approx 6.0\text{e-}1$; $G = 6.674\text{e-}11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

This paper documents the UQFF Learning Assessment Evolution_B module, the ninth in a series of 500+ UQFF code files. Unlike all preceding modules (which target specific astrophysical objects), this is the first **framework-level meta-assessment calculator** in the CP1/CP2/CP3 pipeline. It computes a structured advancement score aggregating three orthogonal metrics: physical regime diversity, dynamical term novelty, and framework scalability. The advancement formula is:

$$\text{advancement} = \frac{\text{diversity_score} + \text{dynamic_score} + \text{scalability_score}}{3.0} \times 100\%$$

The module aggregates parameter inputs from three consecutive UQFF examples: Westerlund 2 stellar wind, Pillars of Creation erosion, and Rings of Relativity lensing modulation — providing a comparative multi-system basis for evaluating framework maturity.

1. Module Context

Source: UQFFLearningAssessment.h (Evolution_B, October 08 2025 revision) Integration target: ziqn233h.cpp (base program, not included) Designed to instantiate as: UQFFLearningAssessment assess; double adv = assess.compute_advancement();

This is the **first CP3 calculator not modelling an astrophysical object** — it models the UQFF framework's own learning progression, providing a quantitative measure of advancement per development session.

2. Core Advancement Formula

$$\text{advancement} = \frac{d + D + s}{3.0} \times 100\%$$

where: - d = diversity_score — number of distinct physical regimes demonstrated (default 3: stellar wind, erosion, lensing) - D = dynamic_score — number of novel dynamic force/field terms introduced (default 3: a_{wind} , $E(t)$ erosion, lensing modulation) - s = scalability_score — adaptability across spatial/temporal scales, normalised to $[0, 1]$ (default 0.8)

Default result:

$$\text{advancement} = \frac{3.0 + 3.0 + 0.8}{3.0} \times 100\% \approx 226.7\%$$

(Values $> 100\%$ indicate multi-regime simultaneous coverage, i.e., framework is demonstrating super-linear progression.)

3. Parameters from Three Prior Examples

3.1 Westerlund 2 (Stellar Wind Regime)

Parameter	Symbol	Default Value	Units
Initial mass	M_{wd2}	$30,000 M_{\odot} = 5.967 \times 10^{34}$	kg
Radius	r_{wd2}	(set per session)	m
Star formation timescale	$\tau_{\text{SF, wd2}}$	3.15×10^{13}	s
Wind density	$\rho_{\text{wind, wd2}}$	1×10^{-20}	kg/m ³
Wind velocity	$v_{\text{wind, wd2}}$	$2,000 \times 10^3$	m/s

3.2 Pillars of Creation (Erosion Regime)

Parameter	Symbol	Default Value	Units
Erosion factor (initial)	E_0	0.3	—
Erosion timescale	τ_{erosion}	3.15×10^{12}	s (~ 0.1 Myr)

Erosion temporal model: $E(t) = E_0 e^{-t/\tau_{\text{erosion}}}$

3.3 Rings of Relativity (Gravitational Lensing Regime)

Parameter	Symbol	Default Value	Units
Einstein radius	r_{rings}	1.54×10^{22}	m
Lensing amplitude factor	L_{factor}	1.2	—
Hubble parameter at z	H_z	2.18×10^{-18}	s ⁻¹

Lensing modulation: $g_{\text{lens}} = U_{g1} \cdot H_z \cdot t + U_{g4}(1 + f_{\text{TRZ}}) \cdot L_{\text{factor}}$

4. Setter Architecture

The original C++ header provides per-variable setters:

```
void setDiversityScore(double v)          / addToVar / subtractFromVar –  
same pattern for all params
```

This enables runtime update of any parameter without reconstructing the object, supporting iterative UQFF refinement workflows.

5. Novel Contribution

1. **First meta-assessment calculator** — evaluates UQFF progression itself, not an astrophysical object
 2. **Three-metric advancement formula** — diversity $\times$ dynamic $\times$ scalability normalised mean
 3. **Multi-example parameter aggregation** — single class absorbs parameters from 3 prior sessions
 4. **Advancement values > 100%** — framework intentionally designed to go beyond binary pass/fail
 5. **Supports quantitative session comparison** — advancement score can be tracked per git commit
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6. CP3 Implementation

Class: UQFFLearningAdvancementCalculator **Method:** compute(dataset) $\rightarrow$ dict **Key output:** advancement_pct (float, %) **Available equations:** wind acceleration, erosion decay, lensing modulation, advancement formula

```
calc = UQFFLearningAdvancementCalculator()  
result = calc.compute({'diversity_score': 3.0, 'dynamic_score': 3.0, 'scalability_score': 3.0})  
# result['advancement_pct']  $\approx$  226.7 %
```

References

- Murphy, D.T. (2025). *UQFF Learning Assessment Evolution_B Module*, UQFFLearningAssessment.h, October 08 2025
- grok_share_8d951e12.txt, Doc 9, lines 2993–3085 (UQFF Learning Assessment Evolution_B)
- Westerlund 2: PAPER_228 (Westerlund2MUGESTellarWindCalculator)
- Pillars of Creation: PAPER_229 (PillarsOfCreationErosionMUGECalculator)
- Rings of Relativity: Prior session UQFFLensingModulationRingsCalculator